

LECTURE 26: MATH1061/7861
TOPICS: PARTIAL ORDER RELATIONS
DATE: WEDNESDAY APRIL 29, 2026

Learning objectives:

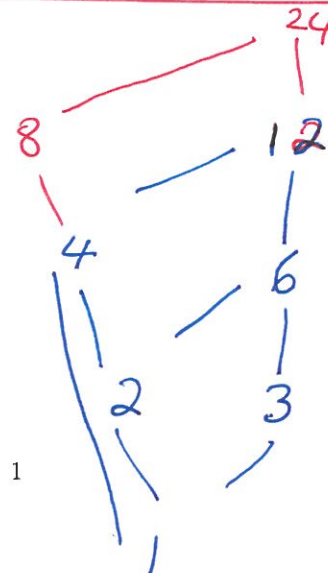
e.g. : $x \leq y$ is a partial order.

- (1) Learn when a binary relation is anti-symmetric.
- (2) Learn when two elements are comparable.
- (3) Learn the definitions of a partial order relation and a total order relation.

Reminder: No Applied classes on Monday (Public Holiday).

Go to any other Applied Class later in the week.

Hasse Diagram a/b:



Q1: Prove that the relation on $\mathbb{Z}_{>0}$ defined by

$$a \tau b \iff a \mid b$$

is a partial order (e.g. reflexive, antisymmetric, transitive).

Reflexive: Indeed. $a = a \times 1$ so $a \mid a$.

Anti-symmetric: We need to show
 $a \mid b$ and $b \mid a \implies a = b$.

Suppose $a \mid b$. This means. $b = am$ $m \in \mathbb{Z}_{>0}$

Suppose $b \mid a$. This means. $a = bK$ $K \in \mathbb{Z}_{>0}$

Putting this together we get:

$$b = am = bKm \implies Km = 1$$

$$\text{because } m, K > 0 \implies m = K = 1$$

$$\implies a = b.$$

Transitive: $a \mid b$ and $b \mid c$ gives

$$b = am \text{ and } c = bK \text{ for } K, m \in \mathbb{Z}_{>0}$$

$$\text{Thus, } c = bK = amK \implies a \mid c$$

Let X be a set.

$\mathcal{P}(X)$

Q2: Let ρ be the relation on $\mathcal{P}(\mathbb{N})$ defined by

$$A \rho B \iff A \subseteq B.$$

(a) Prove ρ is a partial ordering.

Reflexive $A \subseteq A \quad \checkmark$

Anti-symmetric $A \subseteq B \ \& \ B \subseteq A \longrightarrow A = B \quad \checkmark$

Transitive: $A \subseteq B \ \& \ B \subseteq C \longrightarrow A \subseteq C \quad \checkmark$

(b) Draw the Hasse diagram for $\mathcal{P}(\{1\})$, $\mathcal{P}(\{1, 2\})$, $\mathcal{P}(\{1, 2, 3\})$.

$$\mathcal{P}(\{1\}) = \begin{array}{c} \{1\} \\ | \\ \emptyset \end{array}$$

$$\mathcal{P}(\{1, 2\}) = \begin{array}{ccc} & \{1, 2\} & \\ / & & \backslash \\ \{1\} & & \{2\} \\ \backslash & & / \\ & \emptyset & \end{array}$$

$$\mathcal{P}(\{1, 2, 3\}) = \begin{array}{ccccc} & & \{1, 2, 3\} & & \\ & / & | & \backslash & \\ \{1, 2\} & & \{2, 3\} & & \{1, 3\} \\ | & \backslash & / & \backslash & / \\ \{1\} & & \{2\} & & \{3\} \\ \backslash & & | & & / \\ & \emptyset & & & \end{array}$$

Q3: Give some examples of
Partial orderings

Set:

$$\mathbb{Z}_{>0}$$

Relation:

$$a \leq b \text{ if } \underline{\text{lcm}(a,b)} = a$$
$$(\iff b|a)$$

$$\mathbb{R} \times \mathbb{R}$$

$$(x,y) \leq (x',y')$$

$$\underline{x,y,x',y' \in \mathbb{R}}$$

$$\text{iff } x \leq x' \text{ and } y \leq y'$$

Think about
this.